

Monthly Service Prediction for 4G/5G Systems: A Short Time Series-Based Neural Network Solution

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Abstract—Monthly service prediction plays a critical role in the construction planning of 4/5G base stations (BSs) and is a challenging task since the monthly service data is a short time series with a large number of features and insufficient internal logic. In this paper, we develop a short time series prediction framework consisting of deep temporal clustering representation (DTCR) and a decreasing time-difference network (DTD-Net). The former part improves the data quality by dividing the original dataset into multiple mini-datasets with deeper internal logical relationships. The proposed DTD-Net reduces the number of attention operations in traditional time series models by cropping input monthly features block by block and equipping blocks with appropriate attention links, which can solve the problem of overfitting or non-convergence in traditional models. We conduct extensive simulation experiments on the service data of China Mobile to verify the effectiveness of our proposed solution. In particular, an average mean absolute percentage error (MAPE) of 0.126 and 0.12 are achieved for predicting 4G and 5G service indicators, respectively. Moreover, the percentages of the cells with MAPE less than 20% are 79.25% and 84%, respectively. The performance of our proposed solution over traditional methods are also validated via simulations.

Index Terms—Monthly service prediction, short time series, deep temporal clustering representation (DTCR), neural network.

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I. INTRODUCTION

WITH the rapid development of the fifth generation (5G) mobile communication technology, the upgrading of the fourth generation (4G) base stations (BSs) to 5G BSs becomes an important topic in the area of network planning [2], [3], [4]. In particular, whether to deploy a 4G or a 5G BS in a certain district should be carefully considered by the network operator since a correct decision can remarkably improve the user experience and meanwhile save operating costs such as the human resource. The 4G/5G network planning heavily relies on the monthly service data instead of the daily service data since it aims to capture the long-term trend of the network performance. Since most of the service data exhibits temporal correlations, it is possible to predict the future service status based on the historical data, which further assists the network planning.

The network service prediction has been addressed via conventional statistical learning methods in [5], [6], [7]. Li et al. proposed a method based on the Gaussian process regression to predict the data traffic in [5]. Cai et al. exploited spatiotemporal correlations among the mobile data traffic and developed a spatiotemporal Gaussian process Kalman filter to perform the data traffic prediction in [6]. Clemente et al. proposed a Bayesian classifier and a Holt-Winters method to improve the precision of data traffic prediction in [7].

As an advanced machine learning tool, deep learning (DL) has been applied in many scenarios, such as Internet of things [8], virtual reality (VR) [9], [10], computer vision [11], medicine [12], [13] and 6G network design [14]. The potential of DL in the task of time series prediction has also been demonstrated in [15], [16]. In particular, long short-term memory (LSTM) [17], convolutional neural network (CNN) [18], [19], and Transformer [19] become the backbone network structures for time series prediction problems. LSTM is the most commonly used method in current research. The authors of [20] used LSTM to forecast the distribution of cellular traffic in a certain service region, enabling intelligent BS sleeping and lowering energy consumption. A hybrid model comprising double seasonal autoregressive integrated moving average (D-SARIMA) and LSTM was developed to predict the cellular traffic by [21], where D-SARIMA was presented to capture the double seasonality of cellular traffic data, whereas LSTM was used to model the nonlinear residual components of the data. The authors of [22] estimated the

volume of cellular traffic using LSTM, which incorporates min-max normalization and single-exponential smoothing to handle bursts and large fluctuations.

Despite their widespread use, the multiple gate mechanisms of LSTM may introduce considerable temporal complexity and decrease the performance of the entire model. The CNN, which was initially proposed for image processing, has been used as a non-sequential modeling technique for cellular traffic forecasting, and has the advantages of a simpler structure and a faster computation speed over LSTM-based methods. In this regard, the authors of [23] proposed a CNN-based model with 2 channels of traffic input and a residual network (ResNet) structure, each constructed to capture temporal patterns in cellular traffic.

Attention-based models have recently gained popularity in cellular traffic forecasting. The authors of [24] developed an attention-assisted deep learning method for modeling complex patterns in extremely dynamic traffic flows. The self-attention layer was used in a time-embedding block that concatenates both the present and previous states, thus capturing long-term dependencies. Then, to investigate more useful characteristics from the self-attention layer, a feedforward neural network consisting of convolution operation, batch normalization, and rectified linear unit activation was introduced. A wireless network prediction method based on LSTM and attention structures was provided in [25]. For modeling sequence data, a feature extractor made up of LSTM units was used to extract complicated temporal properties. As long sequences may degrade the performance of LSTM-based approaches, the intermediate output of the LSTM was coupled with an attention network. In conclusion, LSTM-based networks have demonstrated remarkable performances in sequential data prediction. However, they are insufficient for modeling extremely lengthy sequences and typically require a considerable amount of training time. CNN-based solutions are easier and faster to train. In addition, attention-based approaches focus on all past input states and treat them differently based on their attention scores, which increases the accuracy of prediction for long-term traffic data.

Although the above models perform well in their own prediction tasks, it is difficult to directly apply them to the prediction of 4G and 5G service. The reasons are as follows: i) Most existing research focuses solely on predicting 4G/5G data traffic, which, however, is only one performance metric for the network service which reflects the user demand. Compared to the mobile traffic prediction, the considered network service prediction involves multiple coupled performance metrics and is hence more challenging. ii) The 5G BSs has only a short official commercialization history and the corresponding monthly time series required by the network planning is very short. Although 4G BSs have been deployed with a longer time, the corresponding monthly time series is still too short for existing prediction models such as LSTM or Transformer. iii) The amount of data is also insufficient due to the limited number of BSs, which leads to the problem of overfitting for traditional attention methods such as the Transformer-based model. iv) Unlike public datasets, our work uses practical datasets whose internal logic is even chaotic within each city. Hence, it becomes quite

difficult to perform service prediction based on the whole dataset.

In this paper, based on the monthly service data of China Mobile in Jiangsu, we consider the prediction of 4G/5G downlink/uplink data traffic and 4G/5G physical downlink/uplink shared channel (PDSCH/PUSCH) physical resource block (PRB) utilization rate. The main contributions of this paper are summarized as follows:

i) Since the practical service data has a rather complicated structure and also a number of flaws caused by incorrect measurement, we develop an efficient data processing strategy to facilitate the network based service prediction. In particular, a data aggregation scheme is proposed to solve the problem of data misalignment caused by multiple 4G/5G cells in the same BS.

ii) A deep temporal clustering representation (DTCR) method is developed to cluster the processed dataset, which firstly extracts information from temporal data using a receptance weighted key value (RWKV) based encoder and then clusters the data using the K-means method. In this way, we implicitly improve the quality of the dataset thus facilitating the subsequent prediction task.

iii) A decreasing time-difference network (DTD-Net) is established based on each clustering output to predict the 4G/5G downlink/uplink data traffic and 4G/5G PUSCH/PDSCH PRB utilization rate. DTD-Net can avoid overfitting caused by a small amount of data and also extract attention relationships month by month, thus achieving a remarkable prediction performance.

iv) Finally, we perform simulation experiments to evaluate the performance of the proposed methods. Specifically, for the prediction of 4G and 5G service data, the mean absolute percentage error (MAPE) achieved by the proposed solution are 0.126 and 0.120, respectively. Moreover, the percentages of the cells with MAPE less than 0.2 are 79.25% and 84%, respectively. Furthermore, ablation experiments are also conducted to verify the necessity of the proposed cascaded solution.

The rest of this paper is organized as follows. Description and processing methods of data are provided in Section II. In Section III, the clustering method DTCR and the prediction model DTD-Net are elaborated. The performance of the proposed solution is demonstrated in Section IV. Finally, Section V concludes this paper.

II. DATA DESCRIPTION AND PREPROCESSING

In this section, we first provide a brief description of the used service data. Considering that the raw data has many flaws, we perform data preprocessing on the raw data which include abnormal data processing, cell data aggregation under the same BS, monthly data alignment, principal component analysis (PCA) and normalization, in order to improve the data quality.

A. Data Description

In this work, we adopt the 4G/5G monthly service data of Jiangsu Province provided by China Mobile from October 2022 to June 2023, where the data of the first eight months

TABLE I
NUMBER OF 4G/5G CELLS PER MONTH IN RAW DATA

Month	4G	5G	Month	4G	5G
Oct-22	611,608	228,213	Mar-23	1247,055	451,463
Nov-22	624,194	243,162	Apr-23	1204,971	266,321
Dec-22	631,559	259,963	May-23	1207,039	429,518
Jan-23	631,197	259,304	Jun-23	581,464	205,538
Feb-23	563,598	258,656			

are used to predict the data of June 2023. The data is collected by the BS every five minutes and transmitted back to China Mobile's network management system. Then, the network management system integrates the received data to form monthly historical data. The data of each month is saved as a separate CSV file, with rows and columns representing the cells and the corresponding features, respectively. The cell is the area served by a base station in a mobile communication system. Specifically, for the raw data, the number of cells in each month is shown in Table I, where each piece of cell-level data has 490 features. The raw data was preprocessed to obtain 27,372 pieces of BS-level data per month, each with 50 features (see the appendix), and finally the number of features was reduced to 9 after PCA. We are interested in the prediction of two kinds of performance metrics: the 4G/5G downlink/uplink data traffic and the 4G/5G PUSCH/PDSCH PRB utilization rate. The former kind of metrics are the features of the original data and the latter kind of metrics can be calculated by the following equation:

$$\text{PRB utilization rate} = \frac{\text{number of occupied PRBs}}{\text{number of available PRBs}}, \quad (1)$$

where both the numerator and the denominator are the features of the original data.

B. Data Preprocessing

Due to measurement errors or data loss, there is a large amount of data with empty and zero values in the raw dataset. The data needs to be carefully cleaned to facilitate the model training. Meanwhile, since each BS serves multiple cells, the data under the same BS needs to be merged and in a suitable way for training. Moreover, there may exist missing data for some months due to factors such as maintenance. Accordingly, we need to align the data by month. Finally, in order to prevent the possible gradient explosion, both PCA and normalization need to be performed on the data.

- (i) Abnormal data processing: Exclude string type features and empty features from the data. Cells with more than half empty features are deleted, along with all other cells belonging to the same BS. Apart from empty values, the zero values should also be carefully handled. Specifically, if the percentage of zeros exceeds 10% for a feature, the feature is deleted. Otherwise, the zeros are filled with the mean value of the corresponding feature. We deleted 41 features, which amounts to 8.36% of the total features. Note that, for the used network data, different features are often correlated. As a result, the removal of a small number of features is unlikely to significantly affect the overall performance.

- (ii) Cell data aggregation under the same BS: Aggregate 4G data and 5G data separately for cells with the same longitudes and latitudes. Some feature values of the cells belonging to the same BS can be directly added, such as the average number of occupied/available PUSCH PRBs, number of terminals, number of E-RABs, number of RRC links and number of users. For features that cannot be directly added, we adopt the following weighted average based aggregation method:

$$R_{\text{agg}} = \sum_{i=1}^{N_s} R_i \cdot \frac{U_i}{U}, \quad (2)$$

where R_{agg} represents the aggregated average user rate, N_s represents the number of cells belonging to the same BS, R_i represents the average user rate in the i -th cell, U_i represents the number of users of the i -th cell, $U = \sum_{i=1}^{N_s} U_i$ represents the total number of users of all cells belonging to the same BS. Finally, we merge the 4G feature dataset and 5G feature dataset for each BS.

- (iii) Monthly data alignment: Owing to the possible maintenance or other reasons, some base stations may have missing data in certain months, which cannot be directly used for training and prediction. Therefore, it is necessary to align the data for each month to ensure that the same BSs are available for each month. To achieve this goal, we sort the BSs based on the latitude and the longitude, and find the BSs whose service data is available for all months.
- (iv) PCA and normalization: As shown in Table IX of the appendix, there are a large number of features which can lead to problems such as overfitting and the curse of dimensionality. In order to facilitate network training, we perform PCA on the collected data to compress the features while retaining 99.5% useful information. In addition, the eigenvalues of different features can vary significantly. In order to maintain a consistent data distribution, we perform Z-score normalization on the tensor data so that the normalized data follows a standard normal distribution. Finally, the dataset after preprocessing is randomly shuffled and split into training and testing sets based on a 4:1 ratio.

III. PROPOSED MONTHLY 4G/5G SERVICE PREDICTION METHOD

The service data depends on economic, geographic or even policy factors, which make the inherent logic quite complicated. Moreover, due to the short deployment time of 5G base stations, data collection is not standardised and other reasons, it is not possible to provide the amount of data that can support large-scale network training. To address this difficulty, we propose to cluster the preprocessed data into several classes such that the data in each class shares a similar logic, instead of directly making predictions on the whole data. Based on the above idea, we propose to use DTCR to classify the data first, and then develop a DTD-Net to predict the data of each class separately. The overall structure is given in Fig. 1. Note that the cells, features, and performance metrics in Section II-A are not directly related to the dataset in this

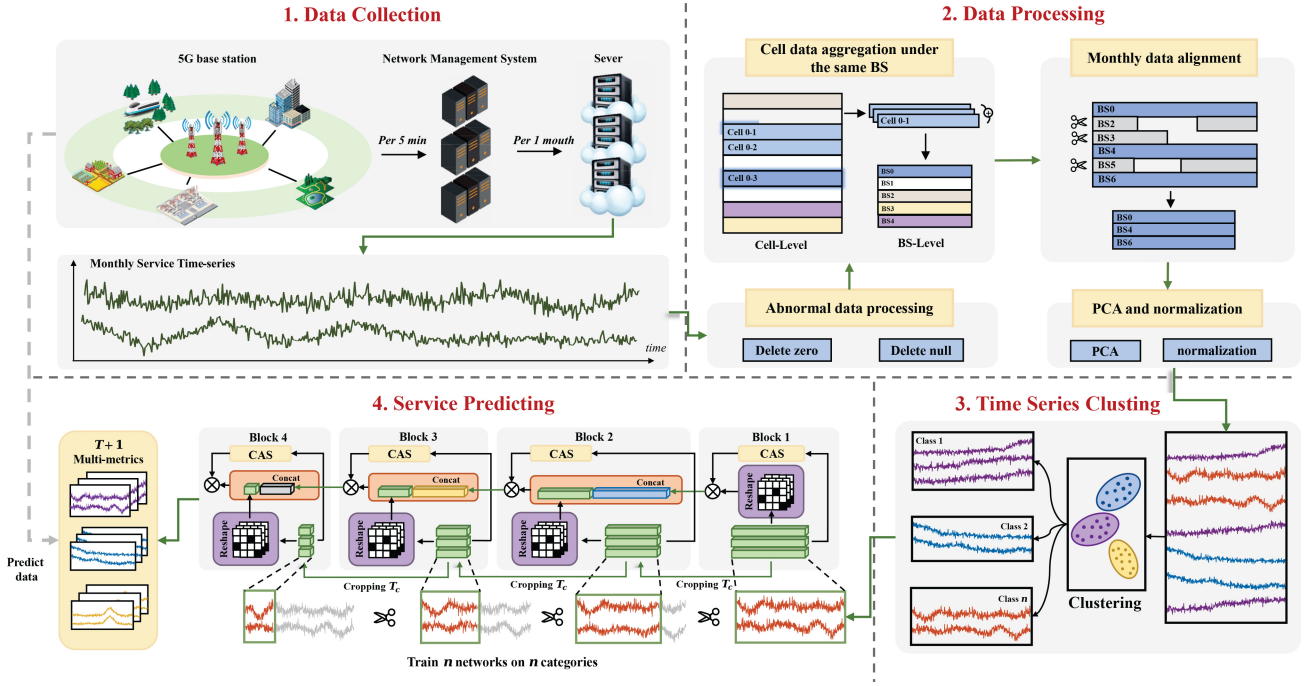


Fig. 1. The overall structure.

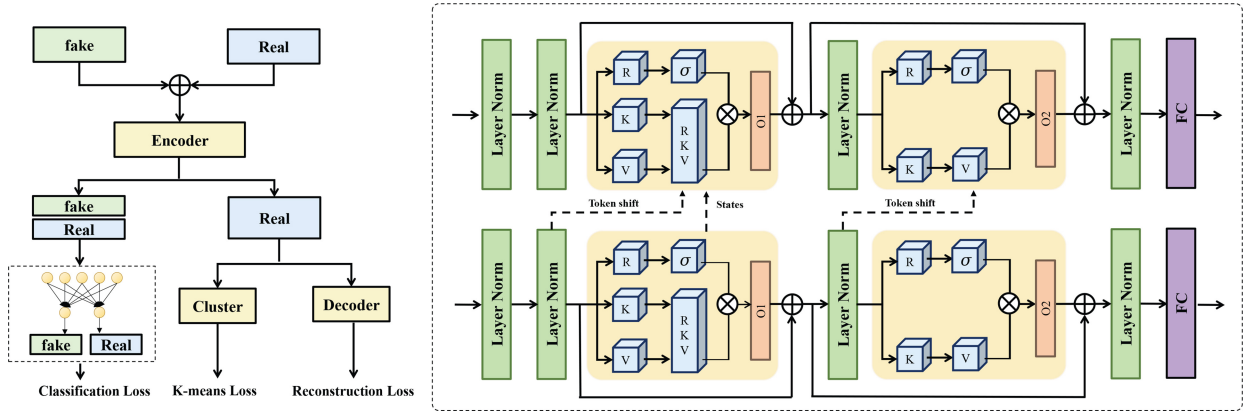


Fig. 2. The general structure of DTCR (left) and the RWKV architecture for time series coding (right).

section since we considered a raw dataset in Section II-A while the preprocessed dataset is considered in this section. Specifically, we aggregate the original cell-level data into BS-level data since the BS-level data offers a clearer representation of regional development trends. Here the data includes both features and performance metrics. We also note that the BS-level features are used as the input of DTDNet in Section III-B and the BS-level performance metrics are used by DTCR in Section III-A.

A. DTCR

Consider a dataset comprising N time series $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$. Each time series $\mathbf{x}_i$ contains T months of performance metrics for the i -th BS, denoted as $\mathbf{x}_i = [x_{i,1}, x_{i,2}, \dots, x_{i,T}]$. Define non-linear mappings $f_{\text{encoder}}(\cdot)$ and $f_{\text{decoder}}(\cdot)$ to represent the encoding and decoding processes, respectively. Then, we have

$$\mathbf{z}_i = f_{\text{encoder}}(\mathbf{x}_i), \quad (3)$$

$$\hat{\mathbf{x}}_i = f_{\text{decoder}}(\mathbf{z}_i), \quad (4)$$

where $\mathbf{z}_i \in \mathbb{R}^m$ and $\hat{\mathbf{x}}_i \in \mathbb{R}^T$ are the output of the encoder and that of the decoder, respectively. The design goal of the encoder is to enhance the utility of the acquired representations for the clustering task, the general structure of DTCR is given in Fig. 2. In this work, we adopt RWKV [26] as the encoder. Compared to the conventional recurrent neural network (RNN) solution, RWKV achieves similar performance as the Transformer while the memory and computation requirements scale linearly with time. The specific structure of RWKV is shown in Fig. 2.

We adopt DTCR since it has superior performance for time series datasets compared to other state-of-the-art deep clustering methods. To improve the performance of clustering, the loss function consists of a reconstruction loss, a classification loss, and a K-means loss. The reconstruction loss ensures that the features learned by the encoder are relevant to the original data. The classification loss makes the features more

discriminative. The K-means loss makes the features more suitable for the clustering task. The reconstruction loss is calculated using the mean squared error (MSE), defined as follows:

$$\mathcal{L}_r = \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2. \quad (5)$$

By minimizing the reconstruction loss, we can learn the features of the original time series data. Since the clustering relies on the capability of the encoder, the better the encoder is trained, the better the learned representation will be. To enhance the capability of the encoder, we use a fake-sample generation strategy and an auxiliary classification task to make the encoder more discriminative. For each original time series, we create a corresponding synthetic sample by randomly shuffling a portion of the series. The auxiliary classification task involves training the encoder to discern whether a given time series is authentic or synthetic. Specifically, the encoder is trained by minimizing the following loss function:

$$\mathcal{L}_c = -\frac{1}{2N} \sum_{i=1}^{2N} \sum_{j=1}^2 y_{i,j} \log \frac{\exp \hat{y}_{i,j}}{\sum_{j=1}^2 \exp(\hat{y}_{i,j})}, \quad (6)$$

where $y_{i,j}$ is the j -th entry of $\mathbf{y}_i$ which is a 2-dimensional one-hot vector representing whether the input is real or synthetic, and $\hat{y}_{i,j}$ denotes the j -th entry of the classification outcome $\hat{\mathbf{y}}_i$. We train using two fully connected layers with 128 and 2 neurons, respectively.

In order to enhance the discriminative nature of the learned features and improve the effectiveness of clustering tasks, we further introduce the K-means loss into the training loss.

In accordance with [27], considering a encoder output matrix $\mathbf{Z} \in \mathbb{R}^{m \times N}$, the K-means objective can be expressed as

$$\mathcal{L}_{\text{K-means}} = \text{Tr}(\mathbf{Z}^T \mathbf{Z}) - \text{Tr}(\mathbf{F}^T \mathbf{Z}^T \mathbf{Z} \mathbf{F}), \quad (7)$$

where $\mathbf{F} \in \mathbb{R}^{N \times k}$ represents the cluster indicator matrix. Fixing the matrix $\mathbf{Z}$ and setting $\mathbf{F}$ to be an arbitrary orthogonal matrix, the minimization of (7) can be converted to a trace maximization problem as follows:

$$\max_{\mathbf{F}} \text{Tr}(\mathbf{F}^T \mathbf{Z}^T \mathbf{Z} \mathbf{F}) \text{ s.t. } \mathbf{F}^T \mathbf{F} = \mathbf{I}. \quad (8)$$

A closed-form solution to $\mathbf{F}$ is given by $\mathbf{F}^* = \text{truncSVD}(\mathbf{Z})$, where $\text{truncSVD}(\mathbf{Z})$ returns the first k singular vectors of $\mathbf{Z}$ achieved by performing a truncated singular value decomposition (SVD).

In the model, $\mathbf{Z}$ represents the features extracted by the encoder, which are dynamic rather than static. We utilize the K-means loss as a regularization term to assist in learning $\mathbf{Z}$ and complement the encoder's learning process. Finally, the overall training loss $\mathcal{L}_{\text{DTCR}}$ for the proposed DTCR model consists of three components as follows:

$$\mathcal{L}_{\text{DTCR}} = \alpha \mathcal{L}_r + \theta \mathcal{L}_c + \lambda \mathcal{L}_{\text{K-means}}, \quad (9)$$

where α and θ are adjustable hyperparameters that determine which type of loss the network prioritizes, λ represents the regularization coefficient. The definitions of $\mathcal{L}_r$, $\mathcal{L}_c$, and $\mathcal{L}_{\text{DTCR}}$ have been shown in (5), (6), and (7), respectively.

Algorithm 1 Training Algorithm of DTCR

Require: the number of clusters K , the number of epochs T , dataset $\mathbf{D}$

- 1: Generate a randomly shuffled series for each sample of dataset $\mathbf{D}$.
- 2: **while** epoch $t = 1$ to T **do**
- 3: Update $\mathbf{Z}$
- 4: Update $\mathbf{F}$
- 5: **end while**
- 6: Obtain the clustering result by applying the K-means method on the encoder output.

Ensure: Clustering result c

The ultimate objective is to minimize the following loss:

$$\min_{\mathbf{Z}, \mathbf{F}} G(\mathbf{Z}) + \frac{\lambda}{2} \left[\text{Tr}(\mathbf{Z}^T \mathbf{Z}) - \text{Tr}(\mathbf{F}^T \mathbf{Z}^T \mathbf{Z} \mathbf{F}) \right], \quad (10)$$

where $G(\mathbf{Z}) = \alpha \mathcal{L}_r(\mathbf{Z}) + \theta \mathcal{L}_c(\mathbf{Z})$. The entire training process of DTCR involves iteratively updating $\mathbf{F}$ and $\mathbf{Z}$. The update strategy for $\mathbf{Z}$ involves fixing $\mathbf{F}$ and then using the standard stochastic gradient descent (SGD) with the gradient given by $\nabla G(\mathbf{Z}) + \lambda \mathbf{Z}(\mathbf{I} - \mathbf{F} \mathbf{F}^T)$. The update strategy for $\mathbf{F}$ involves fixing $\mathbf{Z}$ and updating $\mathbf{F}$ with the analytical solution in (8). The detailed training procedures of DTCR are presented in Algorithm 1.

B. DTD-Net

After applying the proposed DTCR model, we obtain a series of clustered categories, which means that the original dataset is divided into several mini-datasets, each with the same number of features and the same length of time series, but different sizes. The data from each mini-dataset will be input into an individual network with the same structure but different parameters and trained separately. In other words, the number of the individual networks equals that of clustered categories. The network structure for each mini-dataset, which is named as DTD-Net, is shown in Fig. 3. We elaborate the motivation and the details of the network as follows.

For the considered service data, the length of the time series is small, i.e., only nine months of data. Moreover, the data size is also relatively small. Traditional time series networks such as the LSTM structure require a long time series, while attention structures such as the Transformer require a large data size. Therefore, directly applying the existing networks can lead to overfitting or insufficient learning of attention weights between monthly features. To solve this problem, we consider reducing the number of attention operations since performing more attention operations not only requires a higher computational cost, but also requires a longer time series and a larger dataset for training. Furthermore, we find via simulations that the relationship between the feature of interest and auxiliary features is only stronger in the recent few months. Hence, deleting the auxiliary features of the first few months has a very minor impact on predicting these features. To ensure the robustness on different datasets, we need to design a data-driven neural network to learn the impact of

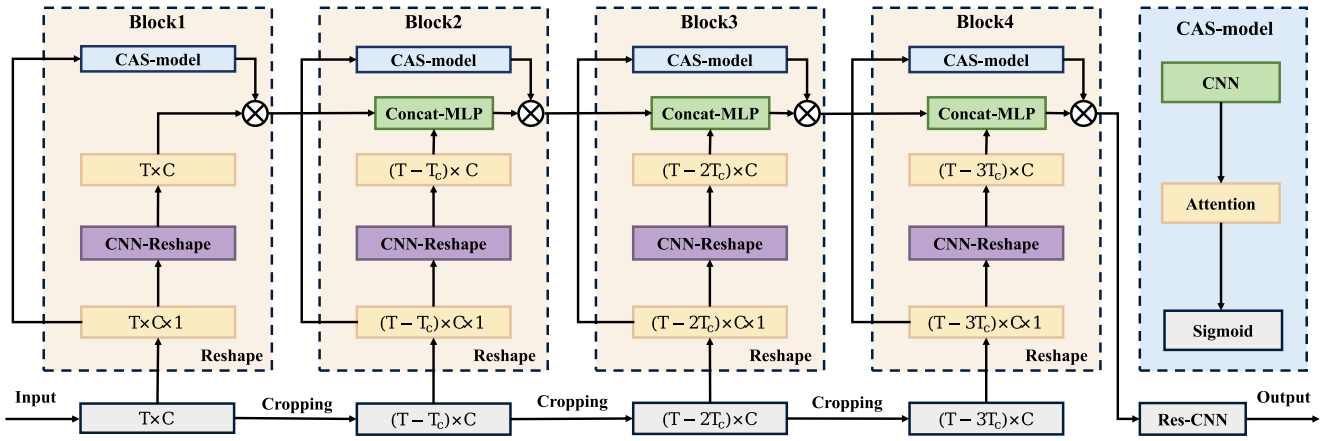


Fig. 3. Schematic diagram of DTD-Net.

input data in different months on the prediction results, thus reducing the number of input months and simplifying the network structure by cropping the number of months.

As shown in Fig. 3, we design several blocks and crop a portion of the monthly data gradually for each block, the length of time series cropped by per block is denoted by hyper-parameter T_c . The number of blocks N_b in the structure depends on the length of time series T and the crop number T_c , which satisfies $N_b = \lfloor T/T_c \rfloor$, where $\lfloor \cdot \rfloor$ indicates the floor operation. C represents the number of features. The smaller the crop number T_c , the finer the data cropping for the same input data by blocks, which means that more detailed relationships between features can be captured while at the cost of higher computational and storage costs.

For our dataset, the length of input time series $T = 8$ and the crop number T_c is set to 2, so the number of blocks N_b in the specific network is 4, with input sizes of these 4 blocks being $T \times C$, $(T - T_c) \times C$, $(T - 2T_c) \times C$, and $(T - 3T_c) \times C$, respectively. These inputs represent all data, data of the latest six months, data of the latest four months and data of the latest two months, respectively. The cropped input data will be input into a convolutional network or a multi-layer perceptron of the block to extract feature information based on the size of the input. The output of the first block is input into a CAS-module, which consists of a CNN, an attention module, and a sigmoid function to adjust feature weights, and then concatenated with the output of the second block. The concatenated features are first input into a multi-layer perceptron, and then input into the second CAS module to adjust feature weights. The same procedure applies to the subsequent blocks. The output of the fourth block is passed through a convolutional module with residual layers to obtain the final prediction result $\hat{y}$. Note that, if new monthly data is supplemented, we can increase the number of blocks in the DTD-Net.

Specifically, the neural network architecture of the backbone and CAS-modules are shown in Table II, respectively. The first two multilayer perceptron (MLP) layers of Block 3 and Block 4 represent the red part of Fig. 3, which is used for feature extraction. The last two MLP layers of Block 3 and Block 4 and the two MLP layers of Block 2 represent the yellow part of Fig. 3, which is used for fusing features between

blocks. Note that CNNs require the input feature map size to be larger than the sliding convolution kernel. This requirement is satisfied in the first two layers, where the input features are sufficiently large. However, in the last two layers, due to the progressive cropping of monthly input features, the feature map size becomes smaller than the convolution kernel. At this stage, we switch from CNN to MLP to reduce computational complexity. All layers are activated by ReLu and normalized with the batch normalization method. All CNNs do not change the size of the input feature map. The attention module in the CAS module adopts two layers with four heads, and the hidden layer dimension is set to 512.

Our prediction requirement is to minimize the MAPE between the predicted features and the ground truth features, but using only the relative error as the loss function can lead to poor convergence. Therefore, we introduce the MSE loss to assist in training. The loss function $\mathcal{L}(\hat{y}; y; \beta)$ is defined as follows:

$$\mathcal{L}(\hat{y}; y; \beta) = \text{MAPE}(\hat{y}, y) + \beta \text{MSE}(\hat{y}, y), \quad (11)$$

where β is a weight hyperparameter used to adjust the priority of MAPE and MSE, $\text{MAPE}(\hat{y}, y)$ and $\text{MSE}(\hat{y}, y)$ are defined as follows:

$$\text{MAPE}(\hat{y}, y) = \frac{1}{B} \sum_{b=1}^B \frac{|\hat{y}_b - y_b|}{y_b}, \quad (12)$$

$$\text{MSE}(\hat{y}, y) = \frac{1}{B} \sum_{b=1}^B (\hat{y}_b - y_b)^2, \quad (13)$$

where B is the batch size. During the network training, the SGD optimizer is used to backpropagate the network parameters to minimize the loss function $\mathcal{L}(\hat{y}; y; \beta)$. The training procedures for the overall network are summarized in Algorithm 2.

IV. SIMULATION RESULTS

In this section, we evaluate the performance of the proposed service prediction solution for 4G/5G systems. We first adopt the Calinski-Harabasz (CH) index [28] as an indicator for evaluating clustering methods, and then we define two indicators,

TABLE II
NEURAL NETWORK ARCHITECTURE OF THE BACKBONE AND CAS-MODULE

	Layer Name	Filters	Heads Num	Layer Num	Input Size	Output Size
Block 1	CNN Layer 1	3×3 kernel	–	–	$T \times C \times 1$	$T \times C \times 1$
	CNN Layer 2	3×3 kernel	–	–	$T \times C \times 1$	$T \times C \times 1$
Block 2	CNN Layer 1	3×3 kernel	–	–	$(T - T_c) \times C \times 1$	$(T - T_c) \times C \times 1$
	CNN Layer 2	3×3 kernel	–	–	$(T - T_c) \times C \times 1$	$(T - T_c) \times C \times 1$
	MLP Layer 1	$(2T - T_c) \times C$	–	–	$(2T - T_c) \times C$	512
	MLP Layer 2	512	–	–	512	$(T - T_c) \times C$
Block 3	MLP Layer 1	$(T - 2T_c) \times C$	–	–	$(T - 2T_c) \times C$	256
	MLP Layer 2	256	–	–	256	$(T - 2T_c) \times C$
	MLP Layer 1	$(2T - 3T_c) \times C$	–	–	$(2T - 3T_c) \times C$	256
	MLP Layer 2	256	–	–	256	$(T - 2T_c) \times C$
Block 4	MLP Layer 1	$(T - 3T_c) \times C$	–	–	$(T - 3T_c) \times C$	128
	MLP Layer 2	128	–	–	128	$(T - 3T_c) \times C$
	MLP Layer 1	$(2T - 5T_c) \times C$	–	–	$(2T - 5T_c) \times C$	64
	MLP Layer 2	64	–	–	64	$(T - 3T_c) \times C$
Res-CNN	Res CNN Layer	3×3 kernel	–	–	$(T - 3T_c) \times C \times 1$	$(T - 3T_c) \times C \times 1$
	MLP Layer	$(T - 3T_c) \times C$	–	–	$(T - 3T_c) \times C$	1
CAS-module 1	CNN Layer 1	3×3 kernel	–	–	$T \times C \times 1$	$T \times C \times 1$
	CNN Layer 2	3×3 kernel	–	–	$T \times C \times 1$	$T \times C \times 1$
	Attention	–	4	2	$T \times C \times 1$	$T \times C \times 1$
CAS-module 2	CNN Layer 1	3×3 kernel	–	–	$(T - T_c) \times C \times 1$	$(T - T_c) \times C \times 1$
	CNN Layer 2	3×3 kernel	–	–	$(T - T_c) \times C \times 1$	$(T - T_c) \times C \times 1$
	Attention	–	4	2	$(T - T_c) \times C \times 1$	$(T - T_c) \times C \times 1$
CAS-module 3	CNN Layer 1	3×3 kernel	–	–	$(T - 2T_c) \times C \times 1$	$(T - 2T_c) \times C \times 1$
	CNN Layer 2	3×3 kernel	–	–	$(T - 2T_c) \times C \times 1$	$(T - 2T_c) \times C \times 1$
	Attention	–	4	2	$(T - 2T_c) \times C \times 1$	$(T - 2T_c) \times C \times 1$
CAS-module 4	CNN Layer 1	3×3 kernel	–	–	$(T - 3T_c) \times C \times 1$	$(T - 3T_c) \times C \times 1$
	CNN Layer 2	3×3 kernel	–	–	$(T - 3T_c) \times C \times 1$	$(T - 3T_c) \times C \times 1$
	Attention	–	4	2	$(T - 3T_c) \times C \times 1$	$(T - 3T_c) \times C \times 1$

Algorithm 2 Training Algorithm of DTD-Net

Require: the number of epochs T_e , batch size B , hyperparameter β , and the number of categories K .

- 1: **while** category $k = 1$ to K **do**
- 2: **while** epoch $t_e = 1$ to T_e **do**
- 3: Select a mini-batch of data from the k -th category $\{(x_b, y_b)\}_{b=1}^B$.
- 4: Inference the predict feature $\{\hat{y}_b\}_{b=1}^B$ by the network shown in Fig. 3.
- 5: Compute the loss $L(\hat{\mathbf{y}}, \mathbf{y}; \beta)$
- 6: Update the k -th set of trainable parameters of DTD-Net via SGD.
- 7: **end while**
- 8: **end while**

Ensure: K sets of network parameters

i.e., MAPE and MAPE₂₀ for evaluating the accuracy of predicting 4/5G service features. Then, we visualize the categories of DTCR clustering and demonstrate the superiority of DTCR clustering performance on our dataset using the Calinski Harabasz index. Furthermore, we study the effect of input time series length on the network. In addition, we compare the predictive performance of our proposed method with classical neural networks. Finally, we also conduct ablation experiments on different clustering methods and neural networks.

A. Performance Metric

We use the following CH index to evaluate the performance of clustering algorithms:

$$CH = \frac{W_{bcs}(N - K)}{W_{wcd}(K - 1)}, \quad (14)$$

where N is the number of samples in the training set and K is the number of categories. W_{bcs} and W_{wcd} represent the between-cluster separation (BCS) and the within-cluster dispersion (WCD), respectively, whose definitions are given by

$$W_{bcs} = \sum_{k=1}^K N_k \|c_k - c\|^2, \quad (15)$$

$$W_{wcd} = \sum_{k=1}^K \sum_{x \in C_k} \|x - c_k\|^2, \quad (16)$$

where N_k is the number of samples in cluster C_k , c_k is the arithmetic mean of the samples in C_k , and c is the arithmetic mean of all the samples. A larger CH index indicates a large BCS and a small WCD, which results in good clustering performance.

We adopt the MAPE metric to evaluate the prediction performance. Meanwhile, we also use MAPE₂₀ to rule out the cases with severe errors. The definition of MAPE is given by (12) and MAPE₂₀ is defined as

$$MAPE_{20} = \frac{1}{N} \sum_{i=1}^N I_{20}[MAPE(\hat{\mathbf{y}}_i, \mathbf{y}_i)], \quad (17)$$

TABLE III
COMPARISON WITH DEEP LEARNING ALGORITHMS WITHOUT CLUSTERING

Network Type	Feature	CNN		LSTM		MLP		Transformer		DTCR+DTD-Net	
		MAPE	MAPE ₂₀	MAPE	MAPE ₂₀	MAPE	MAPE ₂₀	MAPE	MAPE ₂₀	MAPE	MAPE ₂₀
4G	downlink data traffic	0.532	25%	0.573	18%	0.491	27%	0.577	4%	0.140	79%
	uplink data traffic	0.504	19%	0.685	1%	0.459	25%	0.717	1%	0.115	82%
	PDSCH PRB utilization rate	0.379	23%	0.357	5%	0.331	33%	0.633	10%	0.138	79%
	PUSCH PRB utilization rate	0.760	24%	0.882	18%	0.556	24%	0.762	12%	0.111	77%
5G	downlink data traffic	0.767	14%	0.825	15%	0.636	24%	0.685	20%	0.105	83%
	uplink data traffic	0.726	18%	0.657	9%	0.481	35%	0.638	14%	0.126	80%
	PDSCH PRB utilization rate	0.623	30%	0.539	23%	0.448	42%	0.604	11%	0.148	86%
	PUSCH PRB utilization rate	0.528	8%	0.617	20%	0.545	32%	0.930	1%	0.102	87%

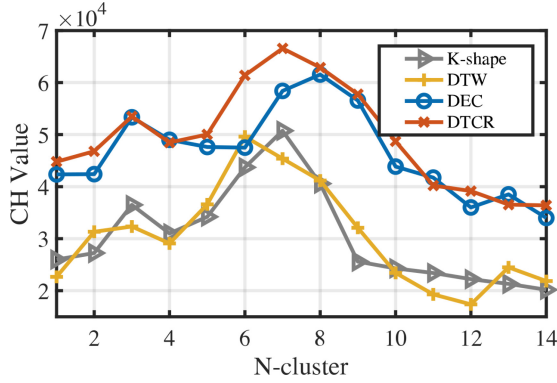


Fig. 4. The CH index of different algorithms versus the number of clusters.

where N is the number of samples in the dataset, $\hat{y}_i$ and y_i are the label and the predicted value of the i -th time-series sample, and the definition of $I_{20}(\cdot)$ is given by

$$I_{20}(x) = \begin{cases} 1, & x \leq 0.2; \\ 0, & x > 0.2. \end{cases} \quad (18)$$

Therefore, MAPE₂₀ denotes the percentage of samples with a MAPE smaller than 0.2.

B. Performance Evaluation

The experiments are run on the Pytorch platform using an Intel Xeon Gold 6330 CPU@2.00GHz, 256-GB RAM and an NVIDIA Geforce RTX 4090 24G GPU. We use the SGD optimiser, with an initial value of 0.001 for the learning rate, and a cosine annealing decay strategy. All weights are initialised with standard normal random variables. The batch size is set to 256. We use RWKV as an encoder to capture the dynamics and multi-scale features of the time series, and the decoder is a single-layer RNN. Gated recurrent units are used in the RNNs [29]. The hyperparameters of the proposed method are α , θ , and β . Since $\alpha + \theta = 1$, we only need to use a two-dimensional grid search to optimize the hyperparameters α and β in the range of (0, 1). The optimized hyperparameters are $\alpha = 0.5$ and $\beta = 0.7$.

The prediction performance of the proposed solution highly depends on the number of clusters. To optimize its value provide the value of the CH index versus the number of clusters in Fig. 4. In order to validate the performance of the improved DTCR, we compare the CH Index of four algorithms for a number of clusters ranging from 2 to 15, including the traditional distance-based clustering algorithm DTW [30], K-shape [31], and the deep clustering method DEC [32]

are considered. It can be observed that the DTCR algorithm achieves the highest CH index when the number of clusters equals 8. For other benchmark schemes, the optimal number is a bit different. However, the highest CH index values of the benchmark methods are all smaller than that of the proposed method. This is due to the fact that DTCR uses the K-means loss as a regularization term, making the extracted features more suitable for cluster characterization. Moreover, using the RWKV encoder can achieve better performance than the traditional RNN while reducing the training overhead. Based on Fig. 4, we get 8 categories of time series after clustering whose sizes are 3261, 4144, 3589, 893, 4727, 1012, 2684, and 7062, respectively.

Network services are influenced by many factors such as city, population, type of area, social activities, etc. Fig. 5 shows four representative categories of 4G/5G data traffic time series, where the downlink and uplink data traffic are added. The time series in Fig. 5(a) presents the traffic trend in a residential area of Suzhou City, Jiangsu Province. This area experiences minor impact from social activities, episodic events, and influx of non-residents, resulting in a relatively stable trend. Different from Fig. 5(a), the time series in Fig. 5(b) shows the traffic trend in a commercial area of Suzhou City, which exhibits a rising trend due to the city's rapidly growing economy. Accordingly, the operator can deploy more BSs in these regions to guarantee the user experience. Fig. 5(c) displays the traffic trend in Fuzimiao, Nanjing, Jiangsu Province. In the fourth month (January 2023), two public holidays, New Year's Day and the Spring Festival, led to an influx of tourists and thus traffic peaks during these months. In general, similar regions exhibit comparable trends. However, there are notable exceptions. As depicted in Fig. 5(d), the demand for 4G and 5G BS services is declining in the commercial buildings in Xuzhou of northern Jiangsu Province. This discrepancy can be due to the difference between the economy development of northern and southern Jiangsu Province and also motivates us to use DTCR for classification instead of only relying on the region type for categorizing.

The eight types of BS data obtain from clustering are separately input into the corresponding DTD-Net for training. In Table III, we compare the proposed DTCR+DTD-Net solution with several commonly used time series prediction algorithm-sin terms of the MAPE and MAPE₂₀ performances. As can be seen, the DTCR+DTD-Net model achieves remarkable prediction accuracy for all considered features. Specifically, the MAPE values are below 0.140 for 4G and 0.148 for 5G, respectively. The MAPE₂₀ values exceed 79% for 4G

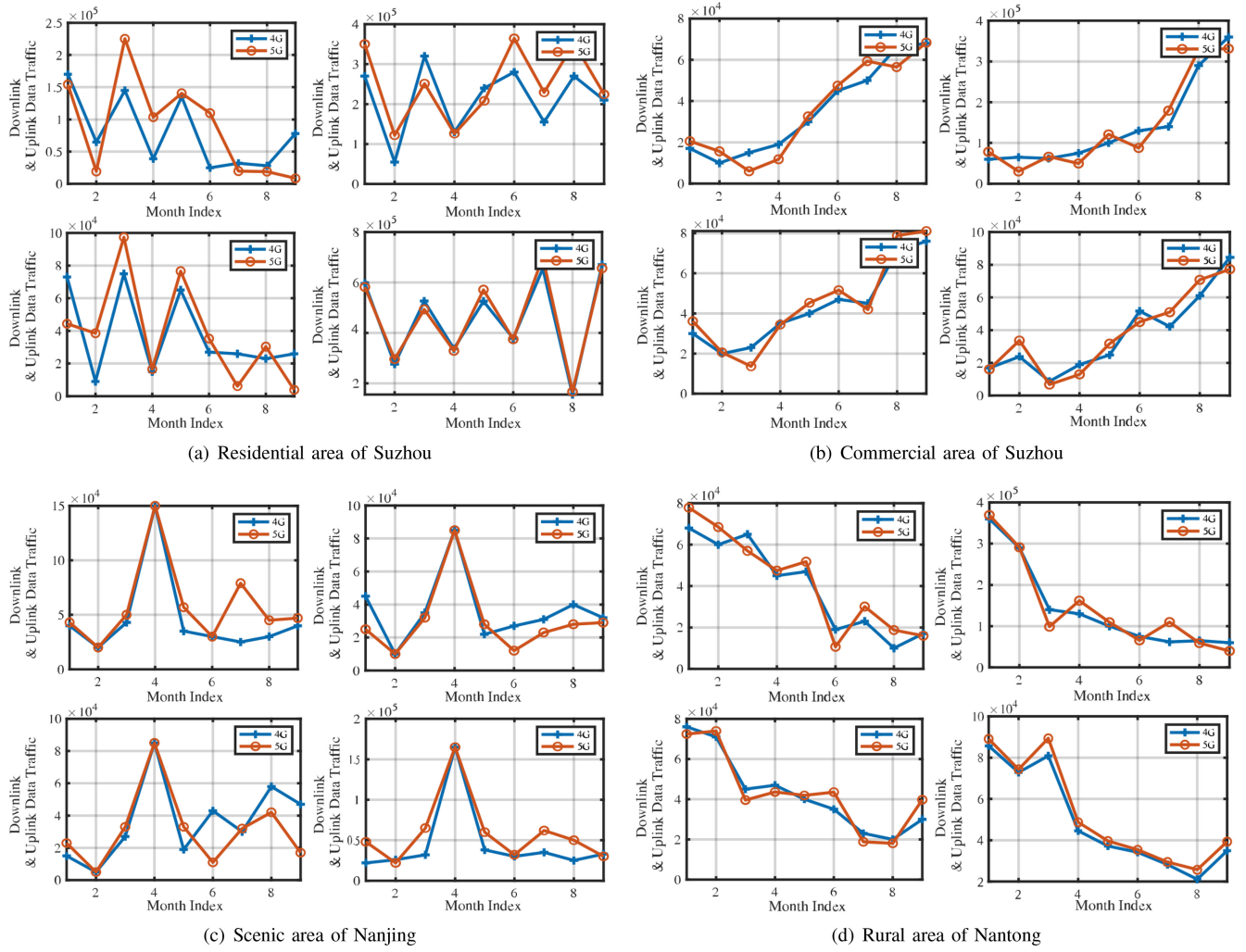


Fig. 5. Four representative categories of 4/5G data traffic time series obtained after DTCT clustering.

and 80% for 5G. On the other hand, traditional models including CNN, LSTM, MLP and the Transformer, exhibit higher MAPE and lower MAPE₂₀ values for both network types. This is because these conventional methods do not perform clustering which fail to model the complex dynamics of network traffic characteristics. It is worthwhile noting that, as will be shown in Section IV-F, the performance of the these conventional networks can be enhanced by performing DTCT based clustering as a preprocessing step. However, the performance advantage of the proposed solution can still be observed (see Table VI). Note that the transformer-based model has a poor performance since it has the overfitting issue due to the used small dataset [19], [33], [34], [35]. More specifically, the Transformer architecture, with its multiple layers of multi-headed self-attention and feed-forward neural networks, offers strong representational power. However, this design also introduces substantial complexity and a large number of parameters. When applied to small datasets, these parameters cannot be sufficiently trained with the limited data, leading to overfitting. The self-attention mechanism excels at capturing global relationships by assessing correlations between positions within a sequence, offering significant flexibility. Yet, in the context of small datasets, this flexibility

can lead the model to overfit local patterns or noises, which further impairs its generalization capability.

C. Impact of Input Time Series Length

To assess the impact of the input time series length on various algorithms, we depict a series of learning curves in Fig. 6 and Fig. 7. We use the data from the latest months when changing the length of time series. For instance, when the length of the input time series equals 2, we use the data in May and April 2023 to predict the service in June 2023. From the figure, we can observe the clear advantage of the DTCT+DTD-Net model over the benchmark schemes as in Table III. Moreover, the MAPE performance improves significantly during the first three or four months for all considered methods. After the fourth month, the rate of the performance improvement for all algorithms decreases which means that an increase in temporal data input results in a minor MAPE reduction. Another notable observation is that the Transformer performs worse than the LSTM. This is because the Transformer architecture requires a substantial amount of data to achieve optimal results. Given the small dataset in this work, it is prone to overfitting and may be outperformed by

TABLE IV
PERFORMANCE COMPARISON USING 4G, 5G, AND 4G+5G DATA

Network Type	Feature	4G		5G		4G+5G	
		MAPE ₂₀	MAPE	MAPE ₂₀	MAPE	MAPE ₂₀	MAPE
4G	downlink data traffic	0.178	73%	0.256	60%	0.140	79%
	uplink data traffic	0.189	77%	0.384	69%	0.115	82%
	PDSCH PRB utilization rate	0.196	72%	0.372	66%	0.138	79%
	PUSCH PRB utilization rate	0.207	72%	0.275	64%	0.111	77%
5G	downlink data traffic	0.340	57%	0.176	78%	0.105	83%
	uplink data traffic	0.297	65%	0.191	75%	0.126	80%
	PDSCH PRB utilization rate	0.390	51%	0.209	77%	0.148	86%
	PUSCH PRB utilization rate	0.348	52%	0.164	79%	0.102	87%

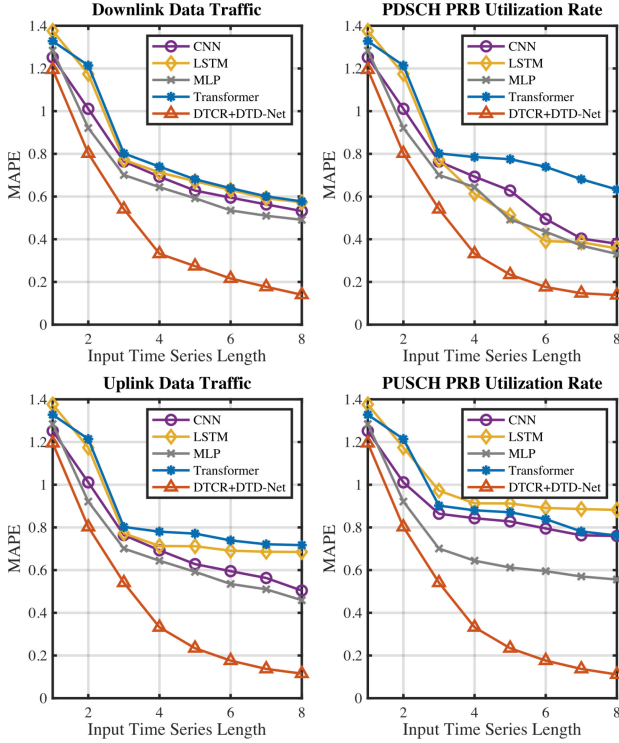


Fig. 6. 4G service prediction performance versus input time series length.

the LSTM. In contrast, LSTM, as a RNN, processes time-series data step by step and retains information incrementally through hidden states, which makes it naturally suited to tasks that involve temporal dependencies, particularly for short time scales [36].

D. Impact of Using Both 4G and 5G Data

To show the impact of using both 4G and 5G data, we conduct experiments to perform service prediction using 4G and 5G data, respectively. The results in Table IV indicate that using either 4G or 5G data alone can lead to an evident performance loss. This is due to the fact that the 4G and 5G services are correlated when the two networks are operating concurrently. The prediction performance can be effectively enhanced by learning the correlation between 4G and 5G data with the proposed solution.

E. Impact of More Historical 4G Data

To show the effect of adopting more historical 4G data, we first utilize 4G and 5G data from February 2023 to May

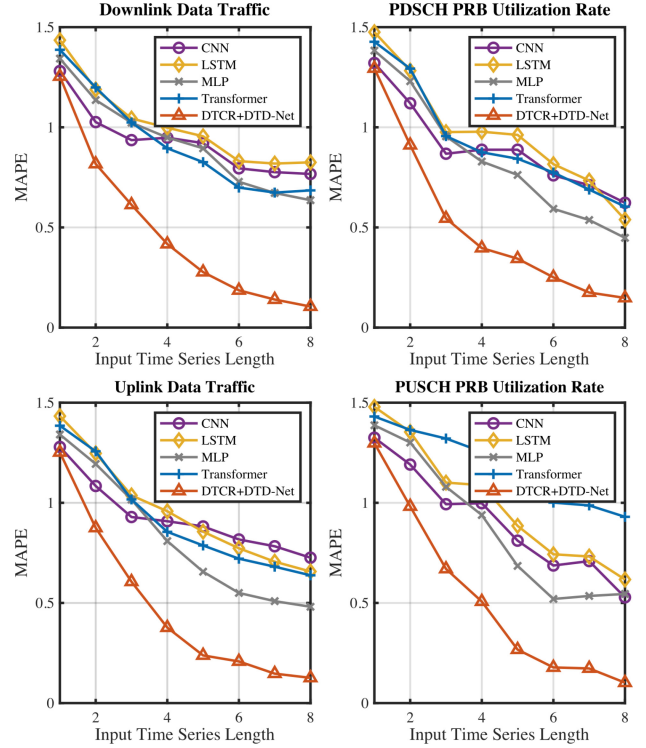


Fig. 7. 5G service prediction performance versus input time series length.

2023 to predict the 5G uplink/downlink data traffic and the PDSCH/PUSCHPRB utilization rate in June 2023. Then, we gradually increase one month of 4G data before February 2023 and evaluate the corresponding prediction performances. The results in Fig. 8 demonstrate that the mean absolute percentage error (MAPE) of different performance metrics decreases as the amount of 4G data increases. Therefore, more historical 4G data indeed enhances the prediction accuracy.

F. Ablation Experiment

In Fig. 4, we have demonstrated the superiority of the DTCR clustering method over other clustering methods in terms of the CH index. In order to demonstrate the necessity and superiority of the DTCR clustering method in predicting service features, we conduct an ablation experiment by using different clustering methods as classifiers and fixing the prediction network with DTD-Net. The specific performance results are shown in Table V.

The results in Table V indicate that the methods utilizing clustering outperform the one without clustering when using

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT CLUSTERING ALGORITHMS COMBINED WITH DTD-Net

Network Type	Feature	DTD-Net		k-shape+DTD-Net		DTW+DTD-Net		DEC+DTD-Net		DTCR+DTD-Net	
		MAPE	MAPE ₂₀	MAPE	MAPE ₂₀	MAPE	MAPE ₂₀	MAPE	MAPE ₂₀	MAPE	MAPE ₂₀
4G	downlink data traffic	0.548	26%	0.376	60%	0.426	38%	0.141	74%	0.140	79%
	uplink data traffic	0.489	20%	0.394	59%	0.130	70%	0.387	51%	0.115	82%
	PDSCH PRB utilization rate	0.396	36%	0.272	66%	0.305	56%	0.194	70%	0.138	79%
	PUSCH PRB utilization rate	0.607	27%	0.175	74%	0.176	74%	0.120	75%	0.111	77%
5G	downlink data traffic	0.740	27%	0.336	63%	0.322	57%	0.213	65%	0.105	83%
	uplink data traffic	0.697	35%	0.391	55%	0.350	53%	0.236	66%	0.126	80%
	PDSCH PRB utilization rate	0.490	41%	0.209	67%	0.252	67%	0.209	62%	0.148	86%
	PUSCH PRB utilization rate	0.548	22%	0.364	59%	0.224	76%	0.246	61%	0.102	87%

TABLE VI
PERFORMANCE COMPARISON OF DEEP LEARNING ALGORITHMS AFTER DTCR CLUSTERING

Network Type	Feature	DTCR+CNN		DTCR+LSTM		DTCR+MLP		DTCR+Transformer		DTCR+DTD-Net	
		MAPE	MAPE ₂₀	MAPE	MAPE ₂₀	MAPE	MAPE ₂₀	MAPE	MAPE ₂₀	MAPE	MAPE ₂₀
4G	downlink data traffic	0.333	58%	0.484	36%	0.269	59%	0.431	29%	0.140	79%
	uplink data traffic	0.343	42%	0.332	27%	0.172	60%	0.324	20%	0.115	82%
	PDSCH PRB utilization rate	0.207	63%	0.289	31%	0.226	57%	0.321	37%	0.138	79%
	PUSCH PRB utilization rate	0.438	45%	0.643	33%	0.435	45%	0.582	28%	0.111	77%
5G	downlink data traffic	0.577	38%	0.626	29%	0.367	47%	0.554	28%	0.105	83%
	uplink data traffic	0.484	36%	0.528	31%	0.317	45%	0.469	36%	0.126	80%
	PDSCH PRB utilization rate	0.356	49%	0.533	44%	0.328	49%	0.418	42%	0.148	86%
	PUSCH PRB utilization rate	0.383	36%	0.515	24%	0.389	65%	0.554	24%	0.102	87%

TABLE VII
ABLATION EXPERIMENT FOR DTCR COMPONENTS

Network Type	Feature	w/o K-means loss		w/o classification loss		both	
		MAPE ₂₀	MAPE	MAPE ₂₀	MAPE	MAPE ₂₀	MAPE
4G	downlink data traffic	0.169	73%	0.246	64%	0.140	79%
	uplink data traffic	0.145	78%	0.364	73%	0.115	82%
	PDSCH PRB utilization rate	0.165	73%	0.354	70%	0.138	79%
	PUSCH PRB utilization rate	0.207	75%	0.242	69%	0.111	77%
5G	downlink data traffic	0.306	58%	0.164	79%	0.105	83%
	uplink data traffic	0.271	68%	0.185	77%	0.126	80%
	PDSCH PRB utilization rate	0.355	54%	0.162	82%	0.148	86%
	PUSCH PRB utilization rate	0.311	54%	0.150	79%	0.102	87%

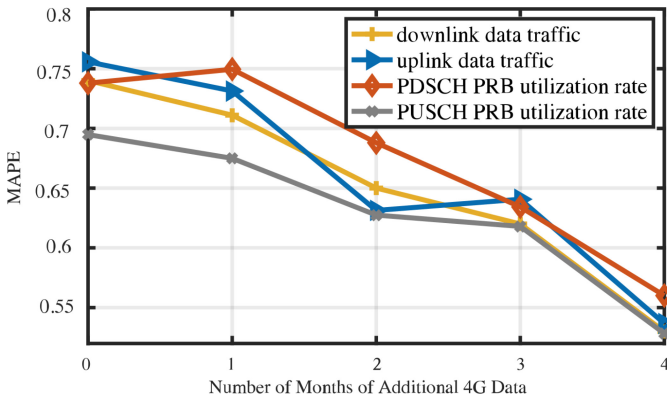


Fig. 8. Impact of more historical 4G data on MAPE of 5G uplink/downlink data traffic and PDSCH/PUSCH PRB utilization rate.

DTD-Net for prediction, which validates the necessity of using clustering methods for the used short time-series data with complex logic. Furthermore, among the methods using clustering, the performance of DTCR is superior to the other three clustering methods. The reasons are given as follows. Although both DEC and DTCR utilize the coding and decoding structure to enable deep clustering, DEC solely employs a basic reconstruction loss without considering the K-means

TABLE VIII
COMPUTATIONAL COMPLEXITY OF DIFFERENT DEEP LEARNING METHODS

Method	Computational Complexity
MLP	$\mathcal{O}(h_m TC)$
CNN	$\mathcal{O}(k^2 TC)$
LSTM	$\mathcal{O}(h_l^2 T + h_l TC)$
Transformer	$\mathcal{O}(C^2 T + T^2 C)$
DTCR+DTD-Net	$\mathcal{O}(TC + KC + \sum_{n=1}^{N_b} (T - nT_c) \times (k^2 C + C^2 + h_m C))$

performance. On the other hand, both K-shape and DTW techniques perform clustering without relying on learning, leading to an evident performance loss.

We also conduct ablation experiments to demonstrate that our proposed DTD-Net outperforms other traditional networks under the same clustering method. The specific performance comparison results are shown in Table VI.

From Table VI, it can be found that the overall performances of Transformer and LSTM are weaker than those of the other three networks. The main reason is that Transformer and LSTM are suitable for long time series prediction and the required data size is typically large, which is not consistent

TABLE IX
SPECIFIC FEATURES

No.	category	Feature Name	No.	category	Feature Name
1	basic information	cell ID	26	CCE related	busy hour number of occupied downlink CCEs
2		BS ID	27		average number of occupied downlink CCEs
3		BS type (4G or 5G)	28		busy hour number of available uplink CCEs
4		latitude	29		average number of available uplink CCEs
5		longitude	30		busy hour number of available downlink CCEs
6		coverage type (village, highway, residential area)	31		average number of available downlink CCEs
7	user related	busy hour uplink user rate	32	data traffic related	busy hour uplink data traffic
8		average uplink user rate	33		average uplink data traffic
9		busy hour downlink user rate	34		busy hour downlink data traffic
10		average downlink user rate	35		average downlink data traffic
11		ARPU (>120)	36	PRB related	busy hour number of occupied PUSCH PRBs
12		ARPU (80-120)	37		average number of occupied PUSCH PRBs
13		ARPU (50-80)	38		busy hour number of occupied PDSCH PRBs
14		ARPU (<50)	39		average number of occupied PDSCH PRBs
15		DOU (>50GB)	40		busy hour number of available PUSCH PRBs
16		DOU (20GB-50GB)	41		average number of available PUSCH PRBs
17		DOU (10GB-20GB)	42		busy hour number of available PDSCH PRBs
18		DOU (5GB-10GB)	43		average number of available PDSCH PRBs
19		DOU (1GB-5GB)	44	others	average number of evolved radio access bearers
20		DOU (300MB-1GB)	45		wireless service utilization efficiency
21		DOU (<300MB)	46		maximum number of radio resource control links
22		total number of users to be charged	47		maximum number of effective radio resource control links
23		total number of users	48		average number of effective radio resource control links
24	CCE related	busy hour number of occupied uplink CCEs	49		average RSRP
25		average number of occupied uplink CCEs	50		total number of RSRP samples

with the considered dataset. Although CNN and MLP have better performances, their network structure is simpler and they can only capture local features, which results in poorer performances than DTD-Net.

Finally, we verify the necessity of using the K-means loss and the classification loss for the DTCR. The results are shown in Table VII where we find that the absence of either K-means loss or classification loss noticeably affects the network performance. This indicates that both components play a crucial role in enhancing the effectiveness of the DTCR framework.

G. Computational Complexity Comparison

Table VIII compares the computational complexity of the DTCR+DTD-Net algorithm with that of the benchmark algorithms. Our method requires a higher computational complexity compared to MLP, CNN, and LSTM. The complexity increase is primarily attributed to the integration of the attention module, which is used to enhance the extraction of global information and also the network performance. The complexity of our method remains lower than that of the Transformer, which is achieved by the proposed time-series clipping method that divides long sequences into shorter sub-sequences, thus avoiding the quadratic complexity associated with the Transformer's global attention mechanism.

Furthermore, Fig. 9 compares the running time of our approach with other methods. The results show that our method needs shorter running time than the Transformer, while requiring longer running time than other approaches. However, our method outperforms all the benchmark schemes in terms of the prediction performance.

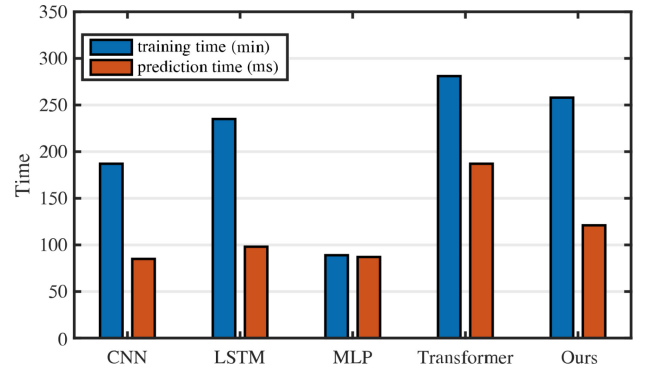


Fig. 9. Comparison of training time and prediction time for all algorithms.

V. CONCLUSION

In this work, we performed monthly service prediction based on China Mobile's 4G/5G service data. We first proposed a data preprocessing approach to facilitate the subsequent prediction task. Then, we further designed a DTCR clustering method using RWKV as the encoder to classify the preprocessed service data, which implicitly improves the quality of the dataset and provides a good foundation for subsequent training of the deep neural networks. Then, a DTD-Net was devised by cropping the input monthly data block by block and fusing the attention relationships within blocks. The number of attention operations is reduced compared to traditional attention based models, which can solve the training problems caused by short time series and insufficient data amount, such as overfitting or non-convergence.

Based on our method, the effectiveness of the method was verified through extensive simulations. By visualizing the DTCR clustering method, we have demonstrated that the time series after DTCR clustering have more internal logic than the ones before clustering, thus we can further perform prediction tasks through data-driven DL networks; We have also demonstrated that the DTCR method outperforms other clustering methods through the comparison of CH index. We explained the rationality of the design of DTD-Net by sequentially inputting monthly data and visualizing the prediction results. Ablation experiments were conducted to demonstrate the necessity of cascading the proposed clustering method and the designed network structure.

There are many interesting research directions in the future regarding this topic, such as how to improve the robustness of models for non-logical mutations in short time series and how to further simplify network structure and reduce training costs for multi-category models.

APPENDIX

SPECIFIC FEATURES DESCRIPTION

The features of our dataset are listed in Table IX, where CCE, ARPU, and DOU are short for control channel element, average revenue per user, and dataflow of usage, respectively. Features are classified into the following types: basic information, user related features, CCE related features, data traffic related features, PRB related features, and others.

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